

FROM TIER-1 TO TIER-0.5: The Rise Of Intelligent Parts And Micro-Interactions

What if the smartest part of a product is not the product itself but the tiny subsystem quietly making decisions and talking back? Designing the future becomes an exercise in coordinating many small, purposeful minds rather than relying on one central core.



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I learned to drive early, earlier than was probably legal, and I learned it from people who treated driving not just as a skill but as a discipline. One lesson, shared almost casually, has stayed with me far longer than any rule or technique.

“Whenever you are about to cross an oncoming vehicle, always, even for a brief microsecond, look into the eyes of the other driver. That glance tells you more than speed, distance, or indicators ever will.”

That fleeting moment is a micro-interaction... a fraction of a second that communicates awareness, confidence, hesitation, and intent. Two humans exchange far more information than any explicit signal could encode and then move on.

Now consider the vehicles we are designing today.

As vehicles become increasingly autonomous, the human on the other side slowly disappears. Which part of the vehicle should I or my autonomous vehicle look at? And how does my vehicle glance back?

It is not a philosophical question. It is a design, engineering, and systems question. And it applies to every product or part we will design for the future.

As intelligence moves away from centralized systems and into edge and zonal computing, i.e., into parts, sub-systems, and embedded electronics, machines are no longer just executing logic. They are making decisions at the edge. And when systems at the edge make decisions, they must also communicate intent, state, and confidence.

A fundamental shift is underway: human-machine interaction (HMI) is moving from the product level to the part level.